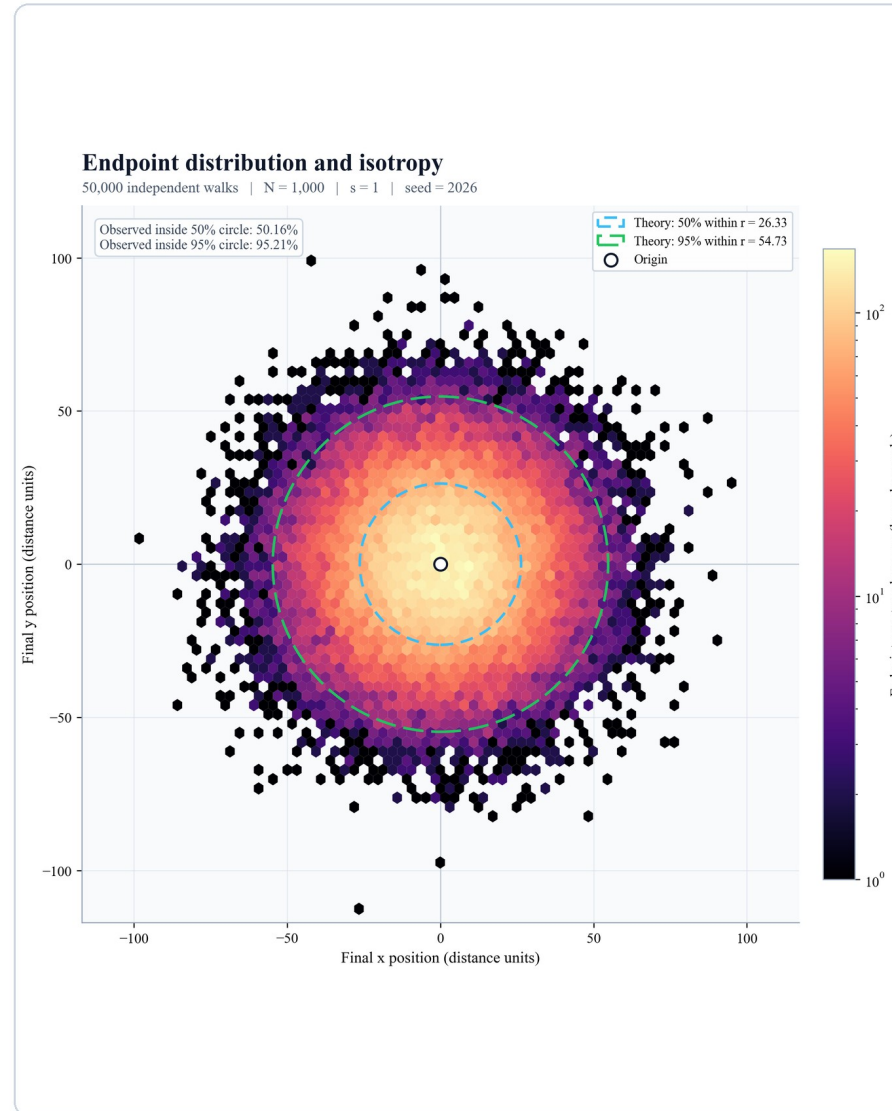
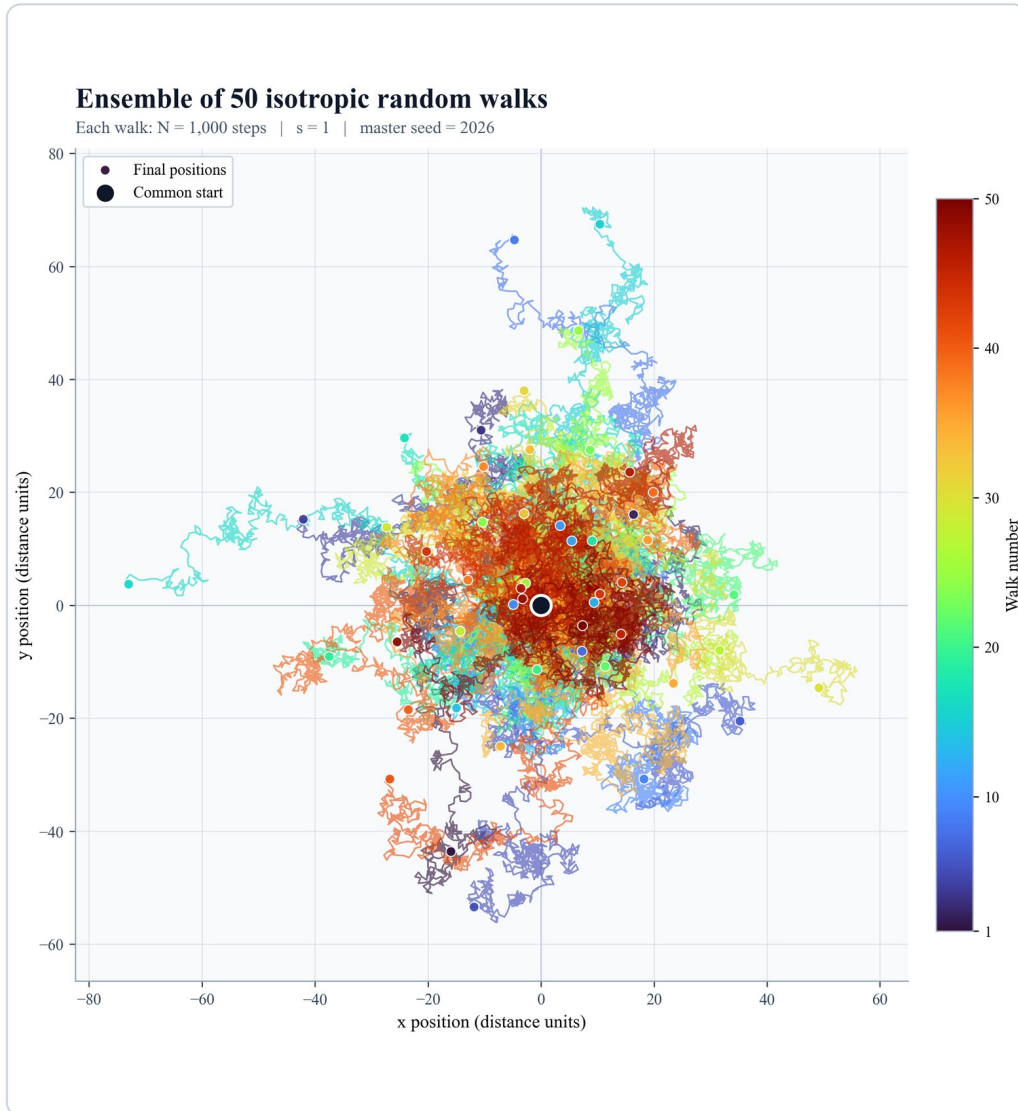


Task 1 · Two-dimensional random walk

Fixed-length steps with uniformly sampled direction produce unbiased isotropic diffusion.



MODEL

$\theta \sim \text{Uniform}$

0 to 2π

step length = s

THEORY

Mean $r^2 = Ns^2$

RMS = $s\sqrt{N}$

SIMULATION

MSD slope

0.9992 ± 0.0031

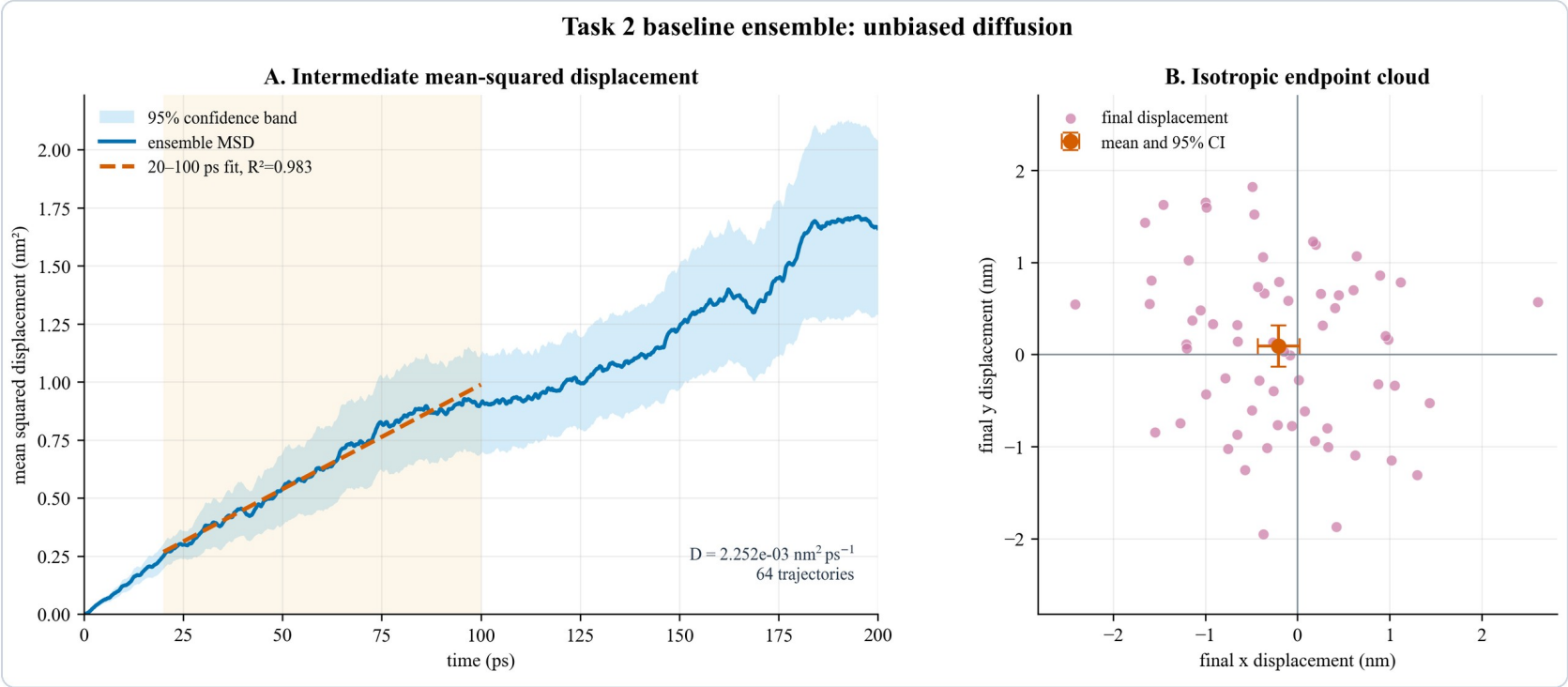
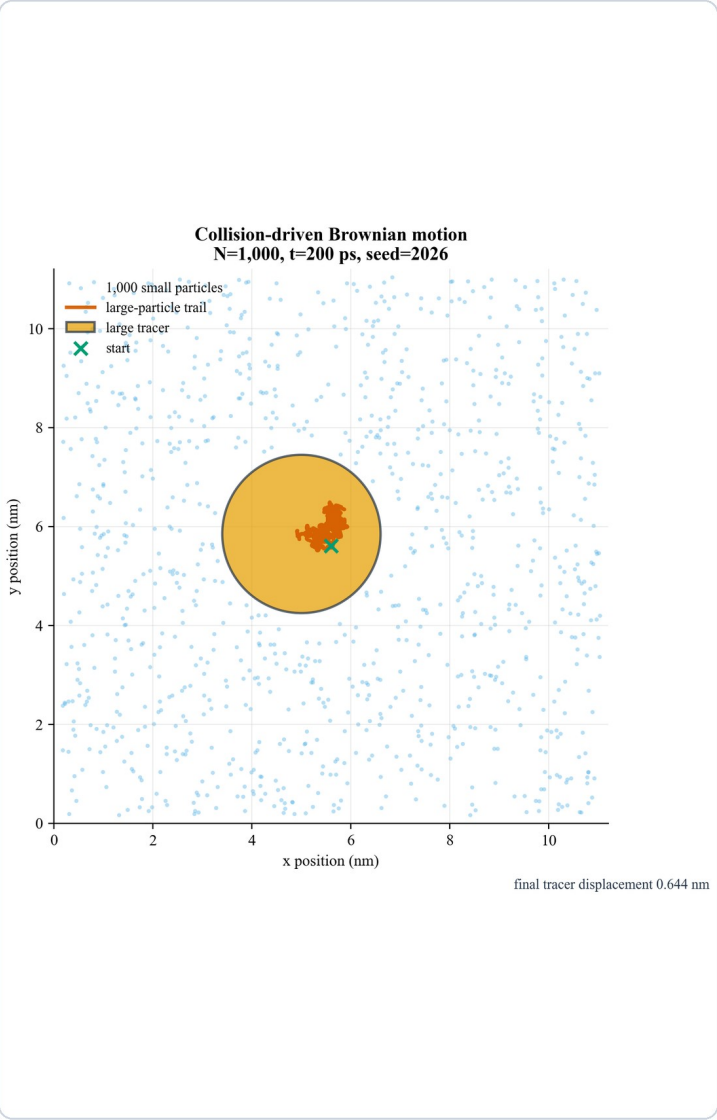
50,000 walks

37/37 CHECKS PASS

50 trajectories show the model; 400,000 independent walks provide the quantitative test.

Task 2 · Collision-driven Brownian motion

A large tracer receives many microscopic impulses, producing an irregular but statistically diffusive path.



MICROSCOPIC MODEL

1,000 particles
200 ps
seed = 2026

DIFFUSION FIT

$R^2 = 0.983$
 $D = 0.00225$
 $\text{nm}^2 \text{ per ps}$

ENSEMBLE EVIDENCE

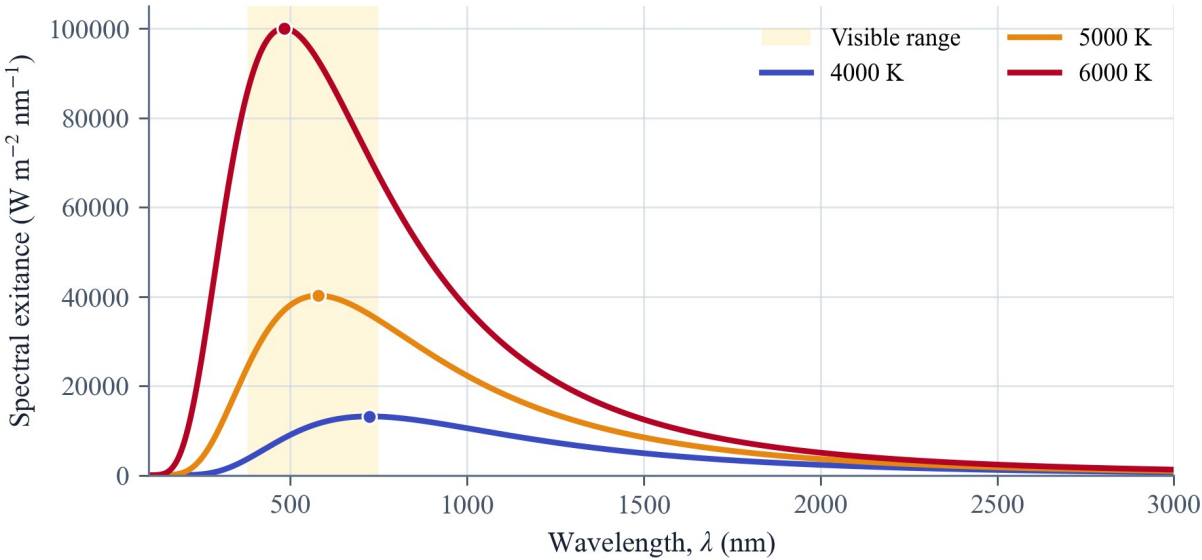
64 trajectories
mean includes zero
unbiased spread

A single irregular path illustrates Brownian motion; the ensemble MSD and endpoint cloud establish diffusion and lack of drift.

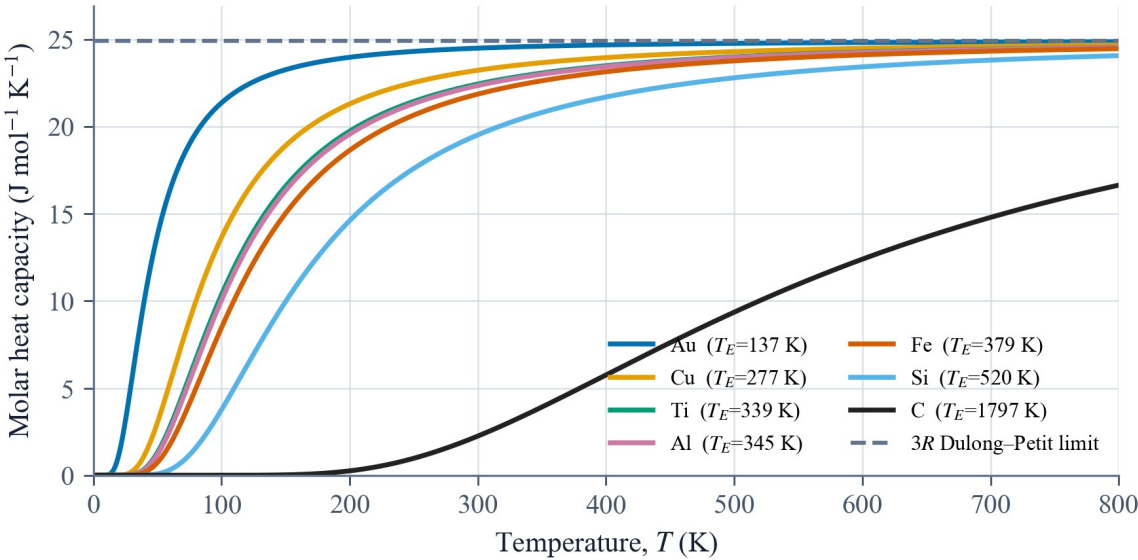
Task 3 — Planck radiation and Einstein heat capacity

Deterministic models • SI units • independently validated limiting laws

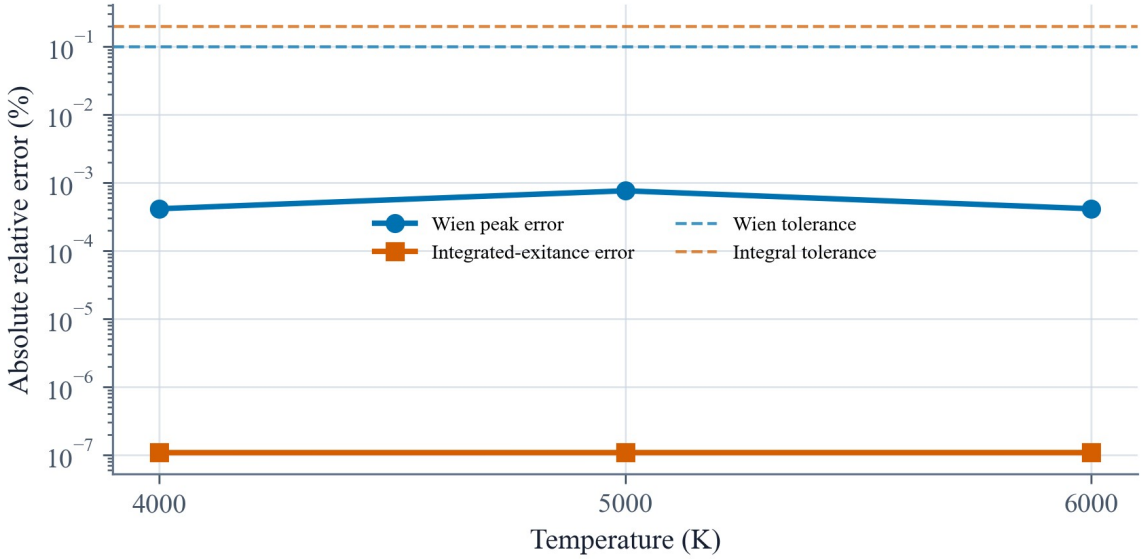
Planck black-body spectra



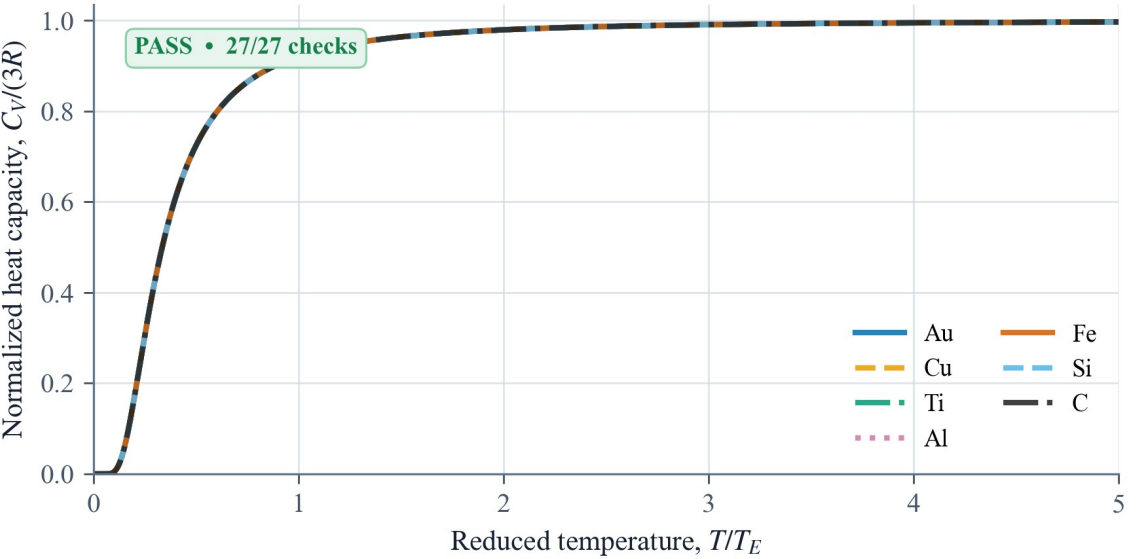
Einstein heat-capacity model



Numerical checks against analytic laws



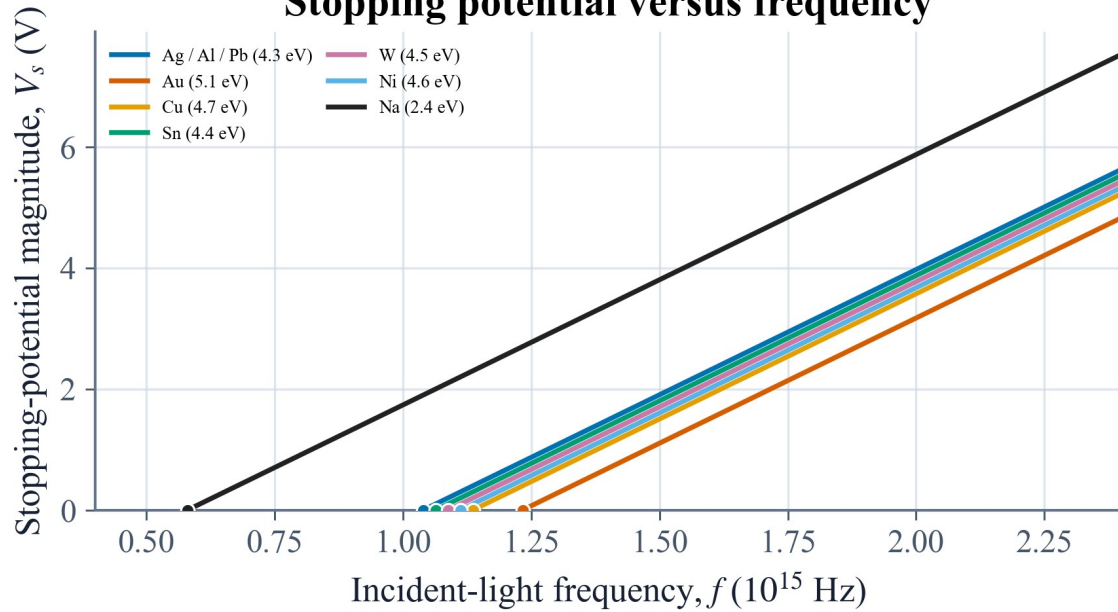
Universal Einstein curve



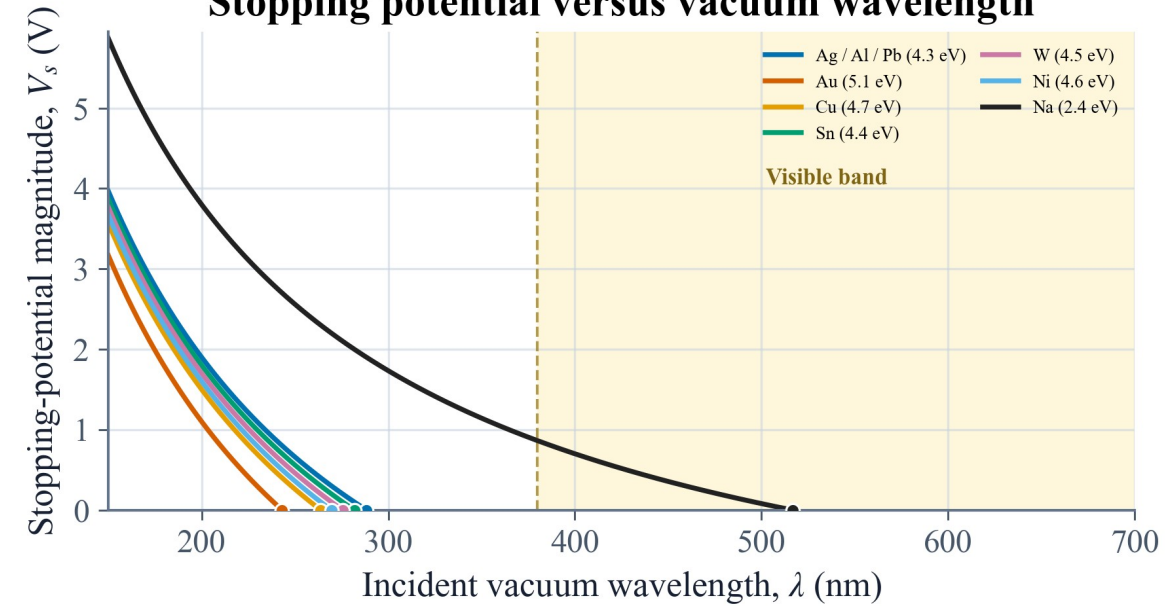
Task 4 — Photoelectric stopping potential

Nine official metals • two equivalent coordinates • physical thresholds shown explicitly

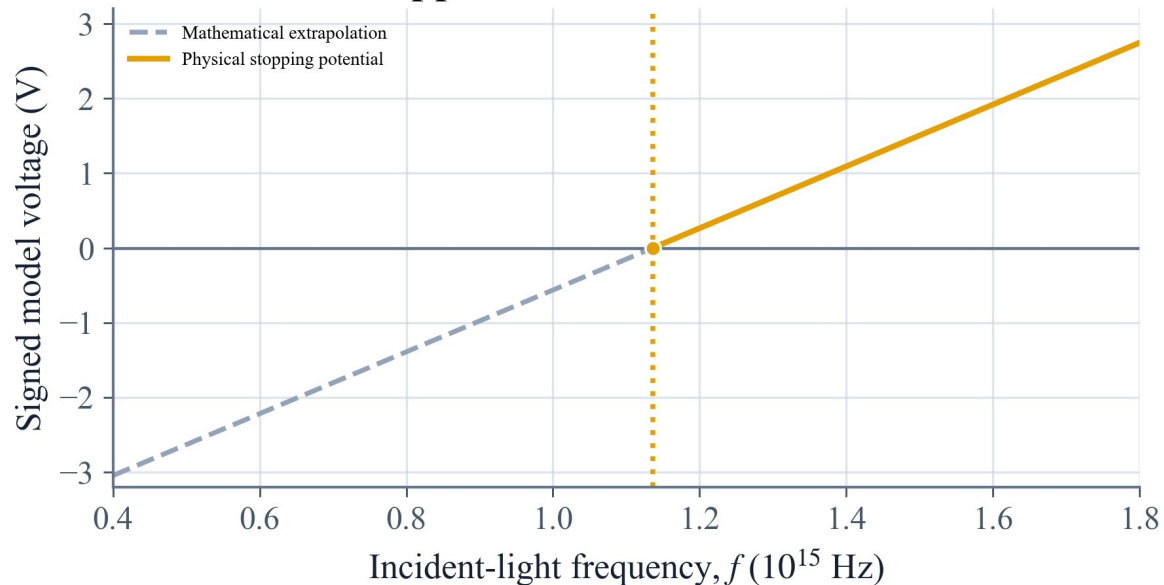
Stopping potential versus frequency



Stopping potential versus vacuum wavelength



Copper threshold: $W = 4.7$ eV



What the simulation shows

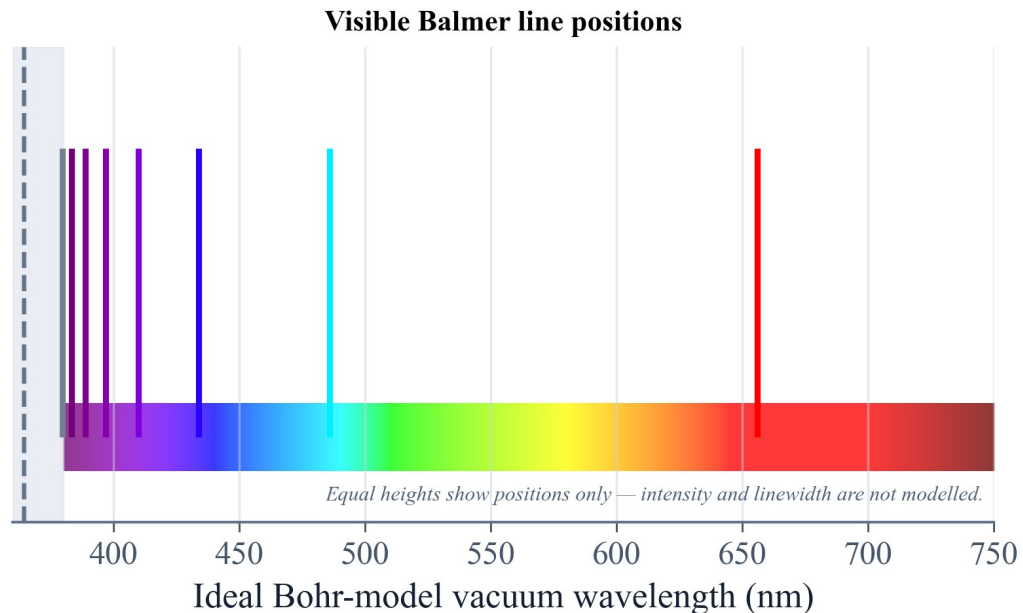
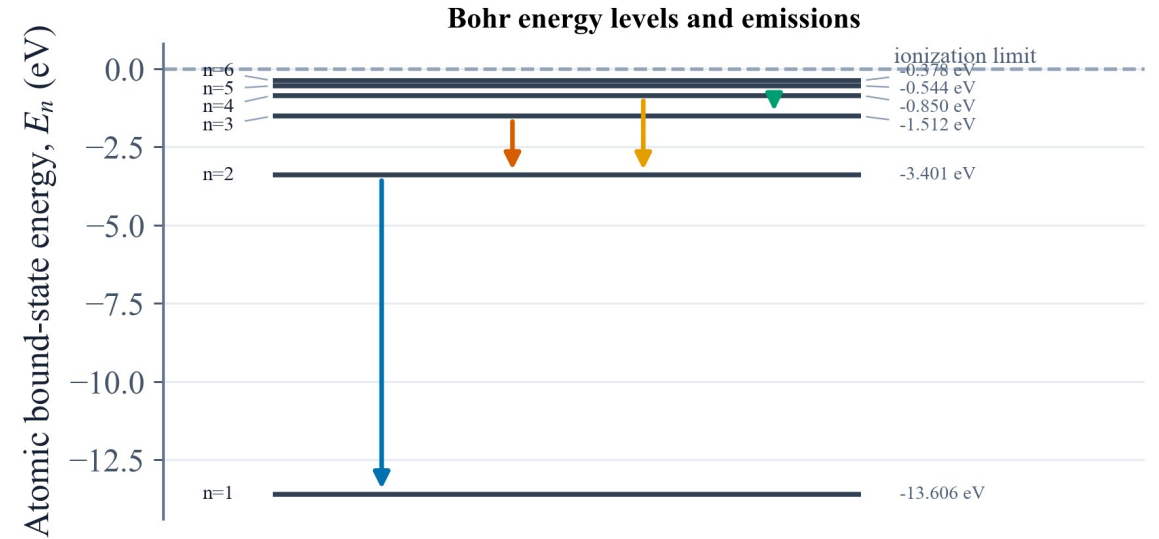
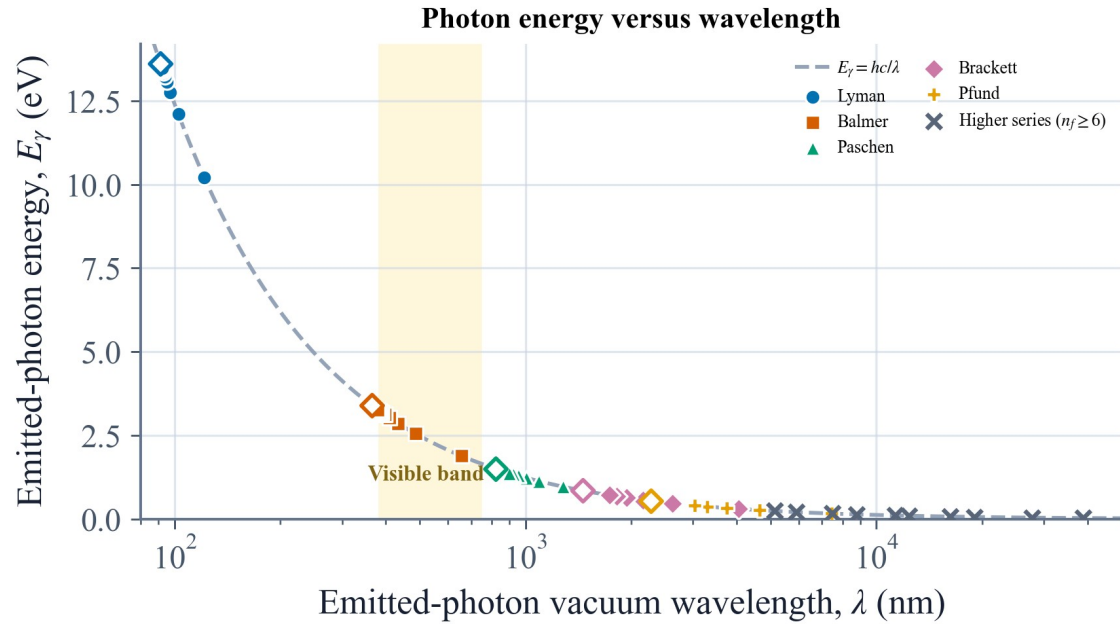
$$eV_s = hf - W$$

- All frequency curves share gradient h/e .
- Higher W raises f_0 and shortens λ_0 .
- Ag, Al and Pb coincide at 4.3 eV.
- Only Na reaches visible light ($\lambda_0 = 516.6$ nm).
- Below threshold there are no photoelectrons.

PASS • 43/43 independent checks

Task 5 — Hydrogen spectrum and the Bohr model

Ten energy levels • forty-five emissions • ultraviolet to infrared



What the model shows

$$E_n = -E_R/n^2$$

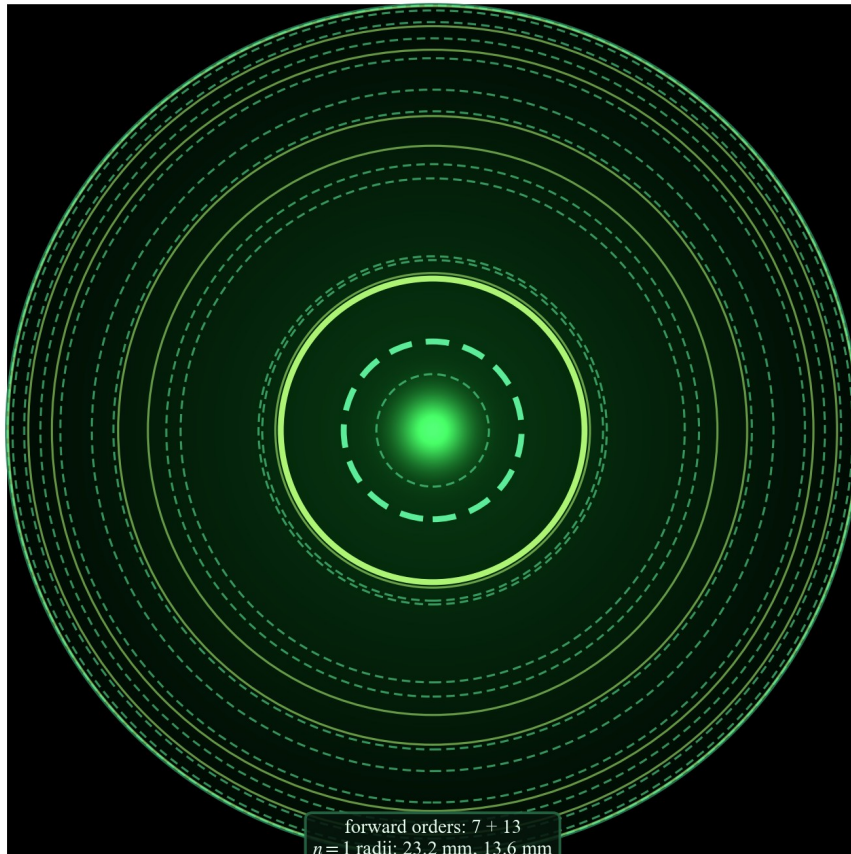
$$E_\gamma = hc/\lambda$$

- 45 downward transitions among $n=1$ --10.
- Integer levels produce discrete photon energies.
- H- α : 656.112 nm in the ideal model.
- Seven declared transitions lie in 380--750 nm.

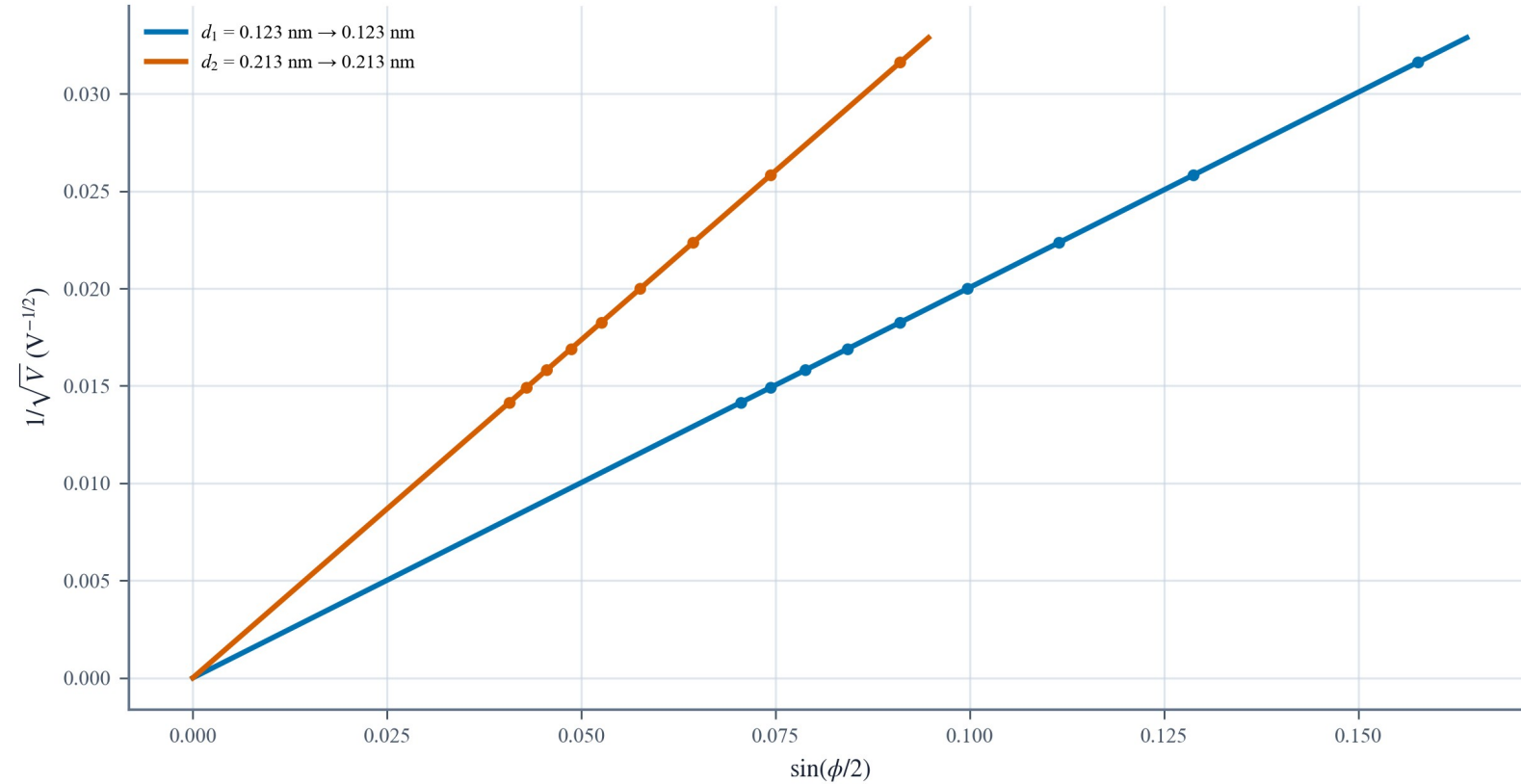
PASS • 30/30 independent checks

Task 6 · Electron diffraction from graphite

Geometric rings at 3 kV



Required straight-line validation



Voltage sweep

1–5 kV
401 values

Order catalogue

11,386 Bragg
7,927 forward

Validation

39/39 checks
passed

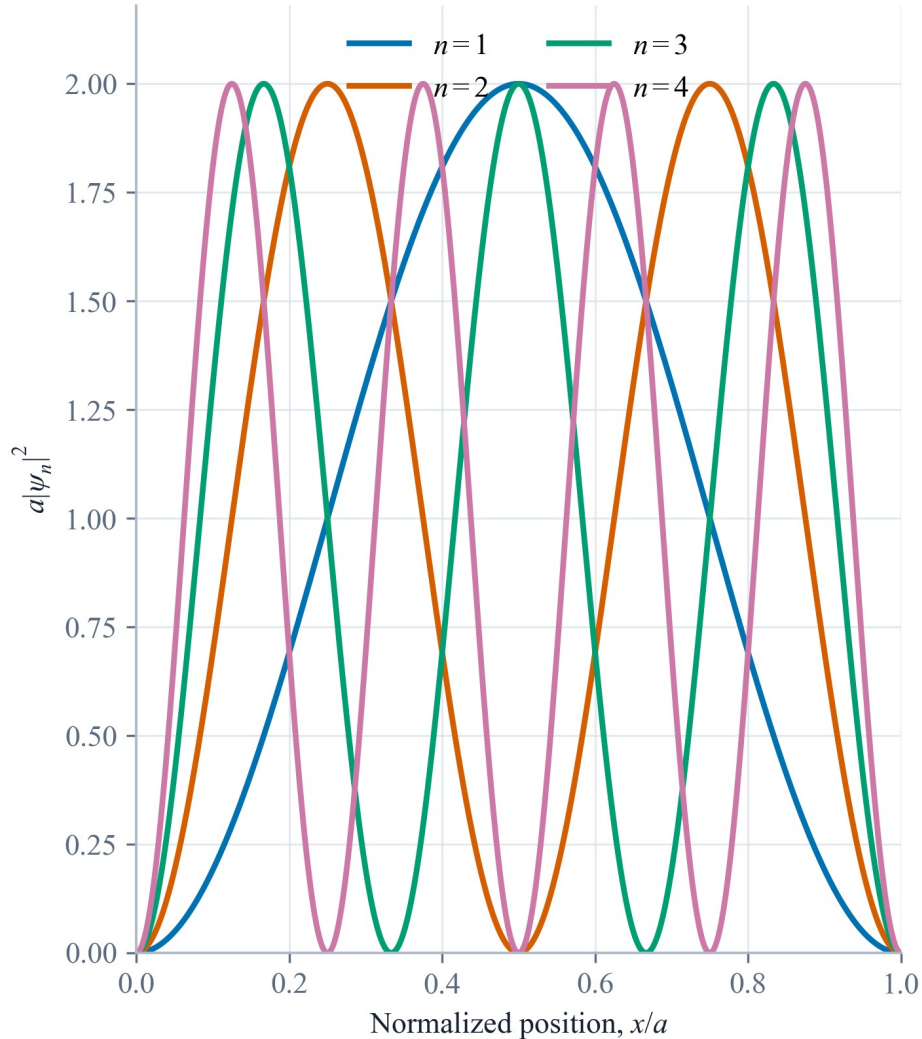
$$\lambda = h/\sqrt{2m_e eV} \quad \cdot \quad 2d\sin\theta = n\lambda \quad \cdot \quad \phi = 2\theta \quad \cdot \quad x = r\sin(2\phi)$$

The model predicts ring positions only. Ring brightness is schematic; no structure factor or detector-response data were supplied.

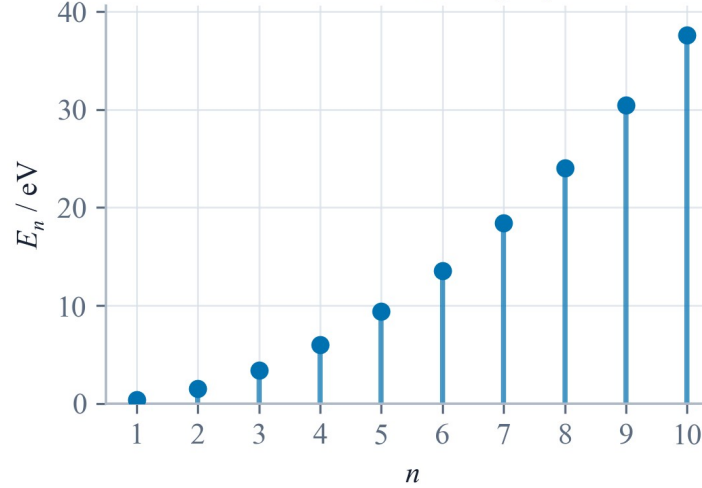
Task 7 · Particle in a one-dimensional box

Infinite-wall boundary conditions create standing waves, discrete energies and unavoidable quantum uncertainty.

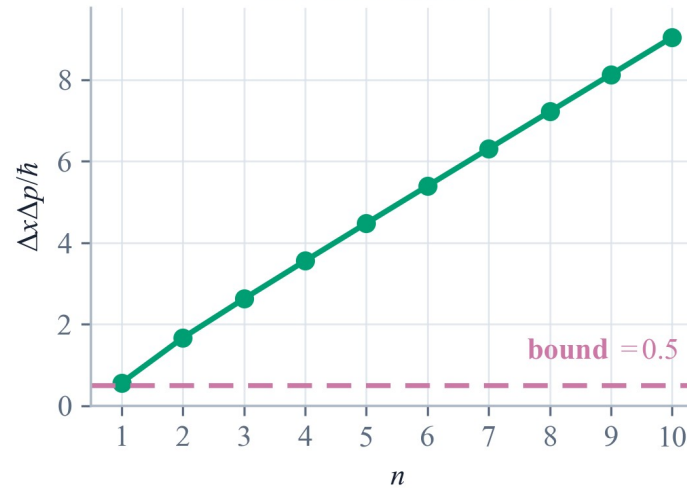
Stationary probability densities



Quantised energy: $E_n/E_1 = n^2$



Heisenberg product



MODEL

$$\psi_n = \sqrt{2/a} \sin(n\pi x/a)$$

$$E_n = n^2 \pi^2 \hbar^2 / (2m_e a^2)$$

Ground energy

0.376030 eV

Ground uncertainty

0.567862 $\hbar$

Numerical check

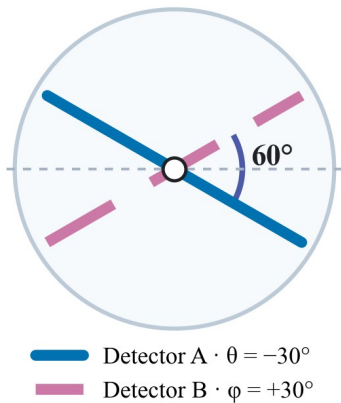
max error 3.21e-05

50/50 CHECKS PASS

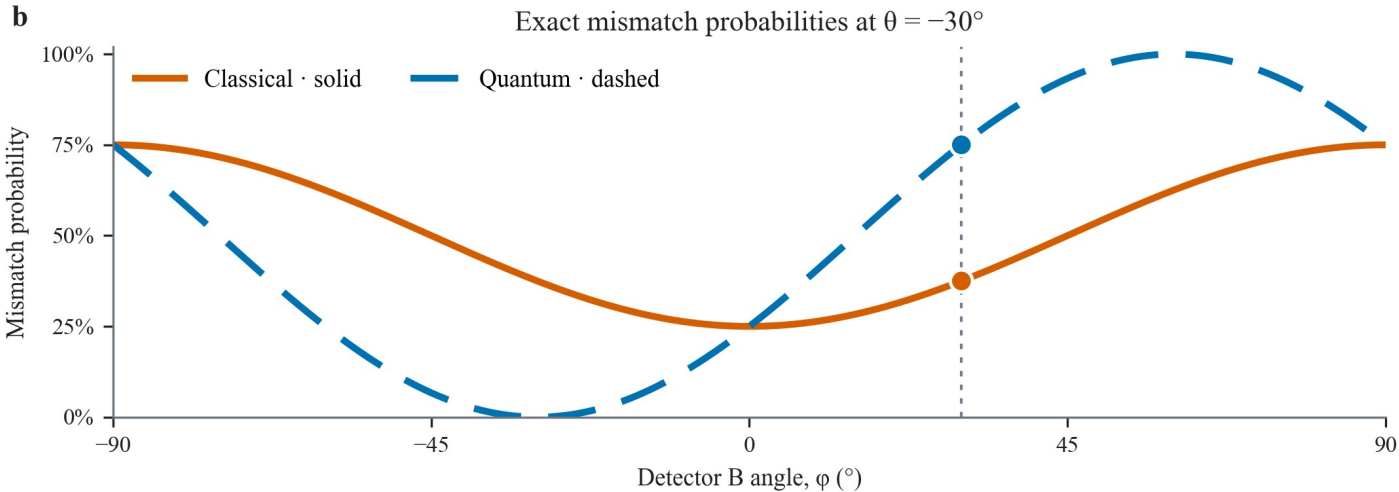
Task 8 · Quantum mismatch calculator

Exact predictions, full angle dependence and one reproducible finite-photon demonstration

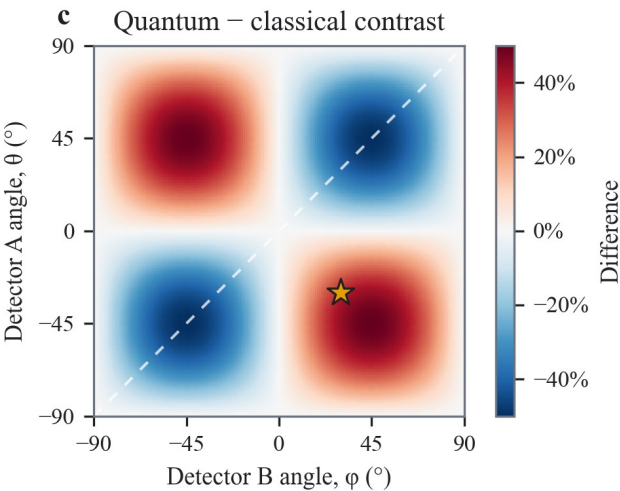
a Detector geometry



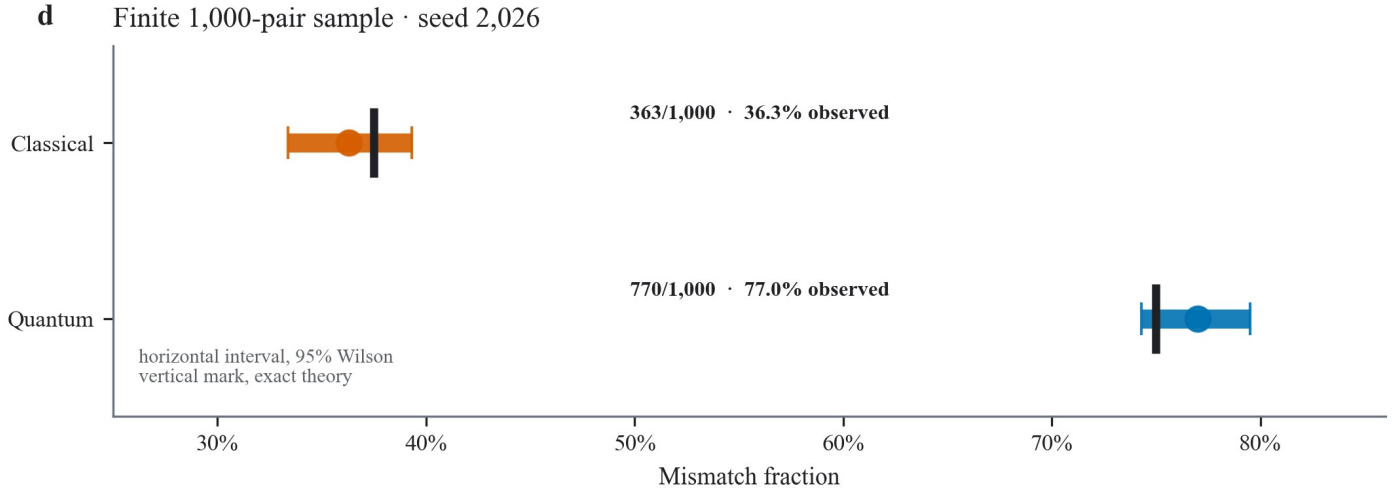
b



c



d

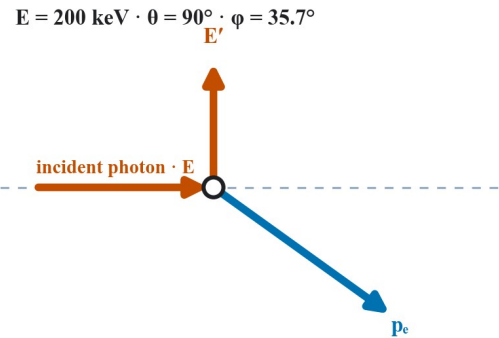


Compton scattering · exact relativistic kinematics

BPhO Computational Challenge 2026 · Task 9 · five official incident energies · stationary free electron

44/44 kinematic checks · 30/30 extension checks

Exact 200 keV collision at $\theta = 90^\circ$ · momentum-vector construction



Fractional wavelength shift

0.39139

$\Delta\lambda = 2.426 \text{ pm}$

Electron recoil speed

0.43419 c

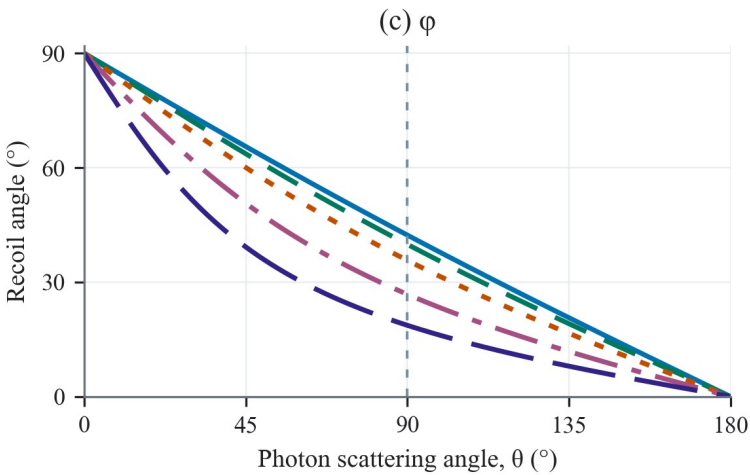
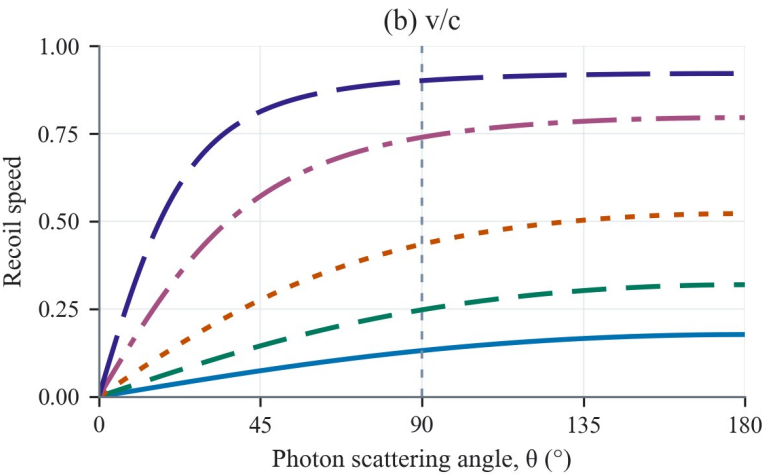
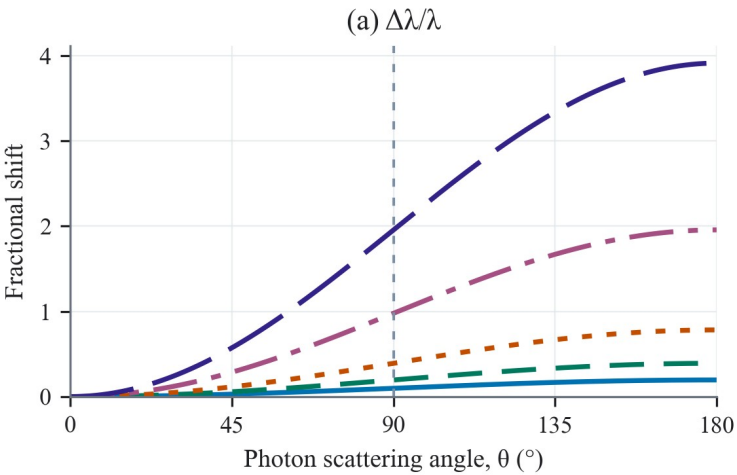
fully relativistic

Electron recoil angle

35.71°

below incident axis

$E' = 143.741 \text{ keV} \cdot K = 56.259 \text{ keV}$



$\theta = 0^\circ$: zero electron momentum and undefined φ (the plots show its 90° limit) · $\theta = 180^\circ$: maximum shift and $\varphi = 0^\circ$

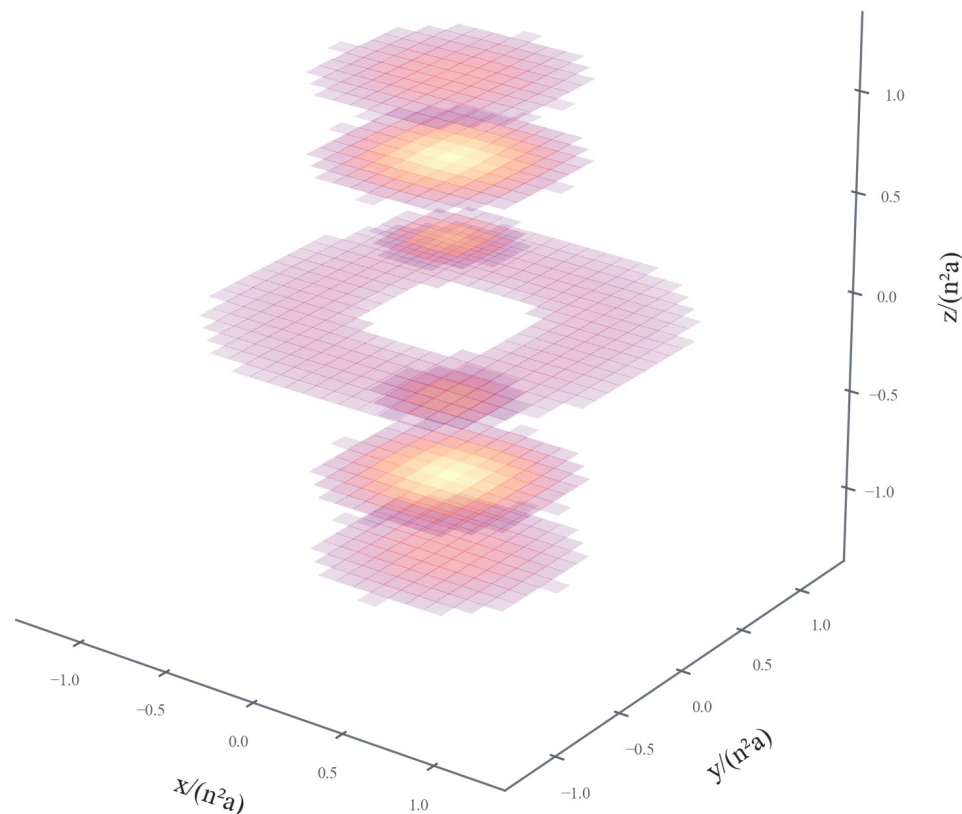
50 keV · solid 100 keV · dashed 200 keV · dotted 500 keV · dash-dot 1000 keV · long dash

Hydrogenic orbitals · normalized 3D density

BPhO Computational Challenge 2026 · Task 10 · one electron, Z protons · complete S–G gallery

22/22 scientific checks · 204 validated states

(a) Required 3D coloured-glass view · hydrogen 3d, m=0



ENERGY

-1.51091 eV

depends on n and Z

99.95% RADIUS

15.13 Å

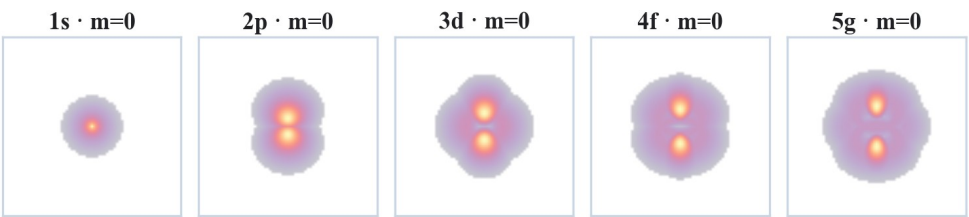
physical H-1 scale

NODES

0 radial · 2 angular

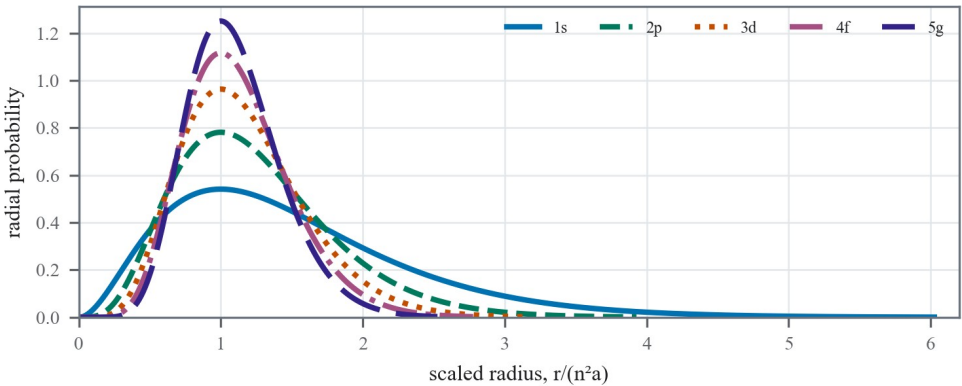
full field normalized

(b) Official S–G family progression



Each view spans 99.95% probability and is normalized only for display.

(c) Radial structure stays distinct from angular shape



Threshold 0.15 changes opacity only · coordinates use standard polar colatitude ϑ and azimuth φ · view rotation is not electron motion